

Exercise 1 of module 1004D on "Theoretical spectroscopy for systems implying periodic boundary conditions"**Operators and commutators**

- a**
- Prove the following commutator relations:

$$[\hat{A}, (\hat{B}, \hat{C})] = [\hat{A}, \hat{B}] + [\hat{A}, \hat{C}] \quad (1)$$

$$[\hat{A}, \hat{B}_1 \hat{B}_2] = [\hat{A}, \hat{B}_1] \hat{B}_2 + \hat{B}_1 [\hat{A}, \hat{B}_2] \quad (2)$$

$$[\hat{A}, [\hat{B}, \hat{C}]] + [\hat{B}, [\hat{C}, \hat{A}]] + [\hat{C}, [\hat{A}, \hat{B}]] = 0 \quad (3)$$

- b**
- Compute the commutators
- $[\hat{x}, \hat{p}_x]$
- ,
- $[\hat{p}_x, \hat{x}^2]$
- , and
- $[\hat{x}, \hat{p}_x^2]$
- with
- $\hat{p}_x = \frac{\hbar}{i} \frac{d}{dx}$
- .

- c**
- Compute the commutators
- $[\hat{l}_i, \hat{l}_j]$
- with
- $i, j = x, y, z$
- .

$$\hat{l}_x = \hat{y} \hat{p}_z - \hat{z} \hat{p}_y, \quad (4)$$

$$\hat{l}_y = \hat{z} \hat{p}_x - \hat{x} \hat{p}_z, \quad (5)$$

$$\hat{l}_z = \hat{x} \hat{p}_y - \hat{y} \hat{p}_x, \quad (6)$$

$$(7)$$

- d**
- Show that
- $\hat{\vec{l}} \times \hat{\vec{l}} = i\hbar \hat{\vec{l}}$
- .

Hermitian operators

An operator is called hermitian if

$$\int \Psi_1^* \hat{O} \Psi_2 dV = \int (\hat{O} \Psi_1)^* \Psi_2 dV, \quad (8)$$

with dV representing integration over all spacial coordinates.

- a** Show that the momentum operator $\hat{p}_i = \frac{\hbar}{i} \frac{\partial}{\partial x_i}$ is hermitian. (Tip: partial integration)
- b** Is the operator $\frac{\partial}{\partial x_i}$ hermitian? Which condition needs to be fulfilled by parameter c of operator $c \frac{\partial^2}{\partial x_i^2}$ so that the operator is hermitian?
- c** Show that hermiticity of operator $\hat{O}$ implies hermiticity of operator $\hat{O}^n$.
- d** Show that eigenvalues of a hermitian operator are real.
- e** Show that eigenfunctions of two different eigenvalues of a hermitian operator are mutually orthogonal.
- f** When is the product of two hermitian operators also hermitian?

- g** Show that the linear combination Ψ of eigenfunctions $\{|i\rangle\}$ of a hermitian operator $\hat{A}$ is an eigenfunction of $\hat{A}$ if the eigenvalues are degenerate:

$$\hat{A}|i\rangle = a_i|i\rangle, \quad |\Psi\rangle = \sum_{i=n}^m c_i|i\rangle, \quad (9)$$

$$\hat{A}|\Psi\rangle = p|\Psi\rangle, \quad a_n = a_{n+1} = a_{n+2} = \dots = a_m = p. \quad (10)$$

Hartree-Fock equations

The Hartree-Fock equations are obtained by variation of the one-electron wave functions (orbitals) $\phi_n(\mathbf{r})$ of the Slater determinant $|\Psi^{\text{SD}}\rangle$,

$$|\Psi^{\text{SD}}\rangle = |\hat{A}(\phi_1(\mathbf{r}_1)\phi_2(\mathbf{r}_2)\phi_3(\mathbf{r}_3)\dots\phi_n(\mathbf{r}_n))\rangle, \quad (11)$$

with $\hat{A}$ indicating the antisymmetrization operator.

- a** Express the expectation value of the molecular electronic Hamilton operator $\hat{H}$,

$$\hat{H} = \sum_i \left(-\frac{1}{2}\nabla_i^2 - \sum_A \frac{Z_A}{r_{iA}}\right) + \sum_{i<j} \frac{1}{r_{ij}}, \quad (12)$$

in terms of one- and two-electron integrals. i indicates electrons, A nuclei, respectively. r_{iA} and r_{ij} indicate the distance between electrons i and j or electron i and atom A .

- b** Take the derivative of the Lagrange functional $\mathcal{L}$,

$$\mathcal{L} = \langle \Psi^{\text{SD}} | \hat{H} | \Psi^{\text{SD}} \rangle + \sum_{ij} \varepsilon_{ij} (\delta_{ij} - \langle \phi_i | \phi_j \rangle) \quad (13)$$

with respect to the one-electron wave functions $\phi_k(\mathbf{r})$ and derive the Hartree-Fock equations by setting the derivative equal to zero.

- c** Show that the energy value of $\mathcal{L}$ remains invariant when applying the unitary transformation,

$$\phi'_i(\mathbf{r}) = \sum_j U_{ij} \phi_j(\mathbf{r}), \quad (14)$$

with $\mathbf{U}$ indicating a unitary matrix and $\phi'_i(\mathbf{r})$ the transformed orbitals.

Koopmans' theorem

Show that the negative orbital energy of occupied orbital ε_i represents an ionization energy if neglecting orbital relaxation.

Roothaan-Hall equations

Derive the Roothaan-Hall equations starting from the canonical Hartree-Fock equations $\hat{F}|\Psi\rangle = \varepsilon|\Psi\rangle$ by inserting the LCAO ansatz and subsequent projection onto the basis functions.